

Convex Optimization for Contraction-Based Control: An Energy-Based Effective Space Approach [★]

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Abstract: In this paper, we propose a computationally efficient and numerically robust contraction metric calculation method for contraction-based control design. Conventional Linear Matrix Inequality (LMI) optimization frameworks for contraction metric calculation suffer from high computational complexity for real-time implementation and scale-invariance issues of the objective function which minimizes the condition number of the contraction metric. To address these issues, we project the contraction metrics onto the trajectory error vector space, leading to a simplified convex optimization framework within a lower-dimensional effective space. Furthermore, energy-based constraints are incorporated to prevent the contraction metrics from collapsing or exploding, thereby maintaining sufficient control authority. In addition, convex input saturation constraints are integrated to handle practical limitations of the actuators. The effectiveness and the feasibility of the proposed method are validated through numerical simulations using the Lorenz system.

Keywords: Contraction theory, convex optimization, input saturation, energy-based control

1. INTRODUCTION

1.1 Motivation

Besides the traditional Lyapunov stability theory, contraction theory offers a interesting perspective on the stability of nonlinear systems. It allows us analyze the system stability by examining the convergence of trajectories. Recent developments in contraction theory based control have introduced computationally efficient methods for obtaining contraction metrics using linear matrix inequalities (LMIs).

Conventional Approaches

- LMI framework for contraction metric calculation

Limitations

- Computational complexity of LMI framework for contraction metric calculation
- No consideration of input saturation issue

To the best of our knowledge, there has been no previous work that projects the contraction constraints onto the error space to reduce the computational complexity of the contraction metric calculation. Moreover, input saturation has not been considered in the conventional contraction theory based control design. Hence, the main objective of this paper is to simplify the contraction metric calculation and to incorporate the input saturation constraint into

a convex optimization framework for contraction metric calculation.

1.2 Contributions

The main contributions of this paper are

- The contraction metrics can be calculated in a lower-dimensional effective space by projecting the contraction constraints onto the trajectory error vector space, which significantly reduces the computational complexity of the LMI framework for contraction metric calculation;
- Scale-invariance issues of the objective function in the conventional LMI framework are resolved by introducing energy-based constraints that prevent the contraction metrics from collapsing or exploding, which ensures minimum required control effort;
- Saturation constraint is incorporated into the convex optimization framework, which handles practical limitations of the actuators; and
- Robust feasibility of the proposed method is enhanced by ... slack variables

2. PRELIMINARIES

Before presenting the proposed method, we first review the contraction theory and related theorems. The detailed explanation of contraction theory can be found in Lohmiller and Slotine (1998); Tsukamoto et al. (2021b).

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2.1 Contraction Property and Contraction Metric

Consider that we have the following deterministic nonlinear system:

$$\frac{d}{dt}\mathbf{x} = \mathbf{f}(\mathbf{x}, t), \quad (1)$$

where $\mathbf{x} \in \mathbb{R}^n$ denotes the state variables and $\mathbf{f} := \mathbf{f}(\mathbf{x}, t)$ denotes the nonlinear system function. Additionally, consider differential dynamics of the system (1) (Kirk, 2004, Chap 4), i.e., $\frac{d}{dt}\delta\mathbf{x} = \frac{\partial \mathbf{f}}{\partial \mathbf{x}}\delta\mathbf{x}$.

According to [...], the system (1) is said to be *contracting*, i.e., all trajectories converge incrementally exponentially to a unique trajectory, if the following condition holds:

$$\frac{d}{dt}\mathbf{M} + 2\text{sym}\left(\mathbf{M}\frac{\partial \mathbf{f}}{\partial \mathbf{x}}\right) \preceq -2\alpha\mathbf{M}, \quad (2)$$

where $\mathbf{M}$ denotes *contraction metric* and $\alpha \in \mathbb{R}_{>0}$ denotes the contraction rate, and $\text{sym}(\mathbf{A}) := \frac{1}{2}(\mathbf{A} + \mathbf{A}^\top)$ Tsukamoto et al. (2021a). The contraction metric $\mathbf{M}$ is symmetric positive definite matrix and satisfies $\bar{m}\mathbf{I}_n \succeq \mathbf{M} \succeq \underline{m}\mathbf{I}_n$ with $\bar{m}, \underline{m} \in \mathbb{R}_{>0}$.

It is notable that the initial conditions are exponentially forgotten in the contracting system Lohmiller and Slotine (1998).

2.2 Perturbed Contracting System

If the system (1) is subjected to bounded disturbances, the system can be expressed as follows:

$$\frac{d}{dt}\mathbf{x} = \mathbf{f}(\mathbf{x}, t) + \mathbf{d}(t), \quad (3)$$

where $\mathbf{d}(t) \in \mathbb{R}^n$ denotes the external disturbances with $\|\mathbf{d}(t)\| \leq \bar{d}$ where $\bar{d} \in \mathbb{R}_{>0}$.

The following theorem states that all solutions of the perturbed contracting system (3) exponentially converge to a bounded ball of the solution of the nominal contracting system (1).

Theorem 1. If the system (1) is contracting with the contraction metric $\mathbf{M}$ which satisfies the condition (2), the path integral $V_l(\mathbf{q}, \delta\mathbf{q}, t) = \int_{\xi_1}^{\xi_2} \|\Theta(\mathbf{q}, t)\delta\mathbf{q}\|$, where $\Theta(\mathbf{q}, t) := \mathbf{M}^{1/2}$, and ξ_1 and ξ_2 are the solutions of (1) and (3), respectively, and $\mathbf{q}$ is the virtual state variable, exponentially converges to a bounded ball. Especially, the following inequality of a bounded error ball of the solutions of (1) and (3) holds:

$$\|\xi_1(t) - \xi_2(t)\| \leq \frac{V_l(0)}{\sqrt{\bar{m}}} \exp(-\alpha t) + \frac{\bar{d}}{\alpha} \sqrt{\chi} (1 - \exp(-\alpha t)), \quad (4)$$

where $\chi := \bar{m}/\underline{m}$ denotes the condition number of $\mathbf{M}$, yielding the steady state error bound of $\frac{\bar{d}}{\alpha}\chi$.

Proof. See the proof of Theorem 2.4 in Tsukamoto et al. (2021b).

3. PROBLEM FORMULATION

Let us consider the following nonlinear system as follows:

$$\frac{d}{dt}\mathbf{x} = \mathbf{f}(\mathbf{x}, t) + \mathbf{B}(\mathbf{x}, t) \text{sat}(\mathbf{u}) + \mathbf{d}(t), \quad (5)$$

where $\mathbf{x} \in \mathbb{R}^n$ denotes the state variables, $\mathbf{u} \in \mathbb{R}^m$ denotes the control inputs, and $\mathbf{d} := \mathbf{d}(t) \in \mathbb{R}^n$ denotes the bounded external disturbances, i.e., $\|\mathbf{d}\| \leq \bar{d}$, and $\mathbf{f} := \mathbf{f}(\mathbf{x}, t)$ and $\mathbf{B} := \mathbf{B}(\mathbf{x}, t)$ denote system functions.

The input saturation function is denoted by $\text{sat}(\cdot)$ and can be defined as any convex function which is physically feasible.

We assume that some desired control input $\mathbf{u}_d \in \mathbb{R}^m$ is pre-designed to achieve the desired trajectory $\mathbf{x}_d \in \mathbb{R}^n$ under ideal conditions, i.e., $\mathbf{d} = 0$ and $\|\mathbf{u}_d\| \leq \bar{u}$ where $\bar{u} \in \mathbb{R}_{>0}$ denotes the maximum allowable input magnitude by simulating the following desired system:

$$\frac{d}{dt}\mathbf{x}_d = \mathbf{f}_d + \mathbf{B}_d\mathbf{u}_d, \quad (6)$$

where $\mathbf{f}_d := \mathbf{f}(\mathbf{x}_d, t)$ and $\mathbf{B}_d := \mathbf{B}(\mathbf{x}_d, t)$.

The error dynamics between the system (5) and the desired system (6) can be expressed as follows:

$$\frac{d}{dt}(\mathbf{x} - \mathbf{x}_d) = \mathbf{f} - \mathbf{f}_d + \mathbf{B} \text{sat}(\mathbf{u}) - \mathbf{B}_d\mathbf{u}_d + \mathbf{d}. \quad (7)$$

Using the state-dependent coefficient (SDC) formulation Dani et al. (2015), i.e., $\mathbb{A} := \int_0^1 \left[\frac{\partial \mathbf{f}}{\partial \mathbf{x}} \right] (c\mathbf{x} + (1-c)\mathbf{x}_d) dc$, the error dynamics (7) can be linearized as follows:

$$\begin{aligned} \frac{d}{dt}(\mathbf{x} - \mathbf{x}_d) &= \underbrace{\mathbf{f} - \mathbf{f}_d + (\mathbf{B} - \mathbf{B}_d)\mathbf{u}_d}_{=\mathbb{A}(\mathbf{x}, \mathbf{x}_d)(\mathbf{x} - \mathbf{x}_d)} \\ &\quad + \mathbf{B}(\mathbf{u} - \mathbf{u}_d) + \mathbf{B}\phi(\mathbf{u}) + \mathbf{d} \\ &= \mathbb{A}(\mathbf{x} - \mathbf{x}_d) + \mathbf{B}(\mathbf{u} - \mathbf{u}_d) + \mathbf{B}\phi(\mathbf{u}) + \mathbf{d} \end{aligned} \quad (8)$$

To simplify the control design, we use the following state feedback control law:

$$\mathbf{u} = \mathbf{u}_d - \mathbf{R}^{-1}\mathbf{B}^\top \mathbf{M}(\mathbf{x} - \mathbf{x}_d), \quad (9)$$

where $\mathbf{R} \in \mathbb{R}^{m \times m}$ denotes a positive definite weighting matrix for the control input. Substituting the control law (9) into the error dynamics (8), we obtain:

$$\frac{d}{dt}(\mathbf{x} - \mathbf{x}_d) = \left(\mathbb{A} - \mathbf{B}\mathbf{R}^{-1}\mathbf{B}^\top \mathbf{M} \right) (\mathbf{x} - \mathbf{x}_d) + \mathbf{B}\phi(\mathbf{u}) + \mathbf{d}, \quad (10)$$

where $\phi(\mathbf{u}) := \text{sat}(\mathbf{u}) - \mathbf{u}$.

Finally, the control objective is to obtain the contraction metric $\mathbf{M}$ in (9) which guarantees the system (5) to be contracting in the presence of input saturation.

4. CONVENTIONAL LMI-APPROACH

In [...], Tsukamoto et al. proposed ConVex optimization-based Steady-state Tracking Error Minimization (CV-STEM) to obtain the contraction metric $\mathbf{M}$ by solving the following LMI optimization problem. Ignoring the input saturation effect, the contraction constraint of the error dynamics (10) can be expressed as follows:

$$\frac{d}{dt}\mathbf{M} + 2\text{sym}(\mathbf{M}\mathbb{A}) - 2\mathbf{M}\mathbf{B}\mathbf{R}^{-1}\mathbf{B}^\top \mathbf{M} \preceq -2\alpha\mathbf{M}. \quad (11)$$

Using the variable transformation $\mathbf{M} =: \mathbf{W}^{-1}$, we have the following LMI optimization problem:

$$\begin{aligned} &\min_{\mathbf{W}, \chi, \nu} \chi + \lambda\nu \\ \text{s.t. } &\left\{ \begin{aligned} -\frac{d}{dt}\overline{\mathbf{W}} + 2\text{sym}(\mathbb{A}\overline{\mathbf{W}}) - 2\nu\mathbf{B}\mathbf{R}^{-1}\mathbf{B}^\top &\preceq -2\alpha\overline{\mathbf{W}}, \\ \mathbf{I}_n &\preceq \overline{\mathbf{W}} \preceq \chi\mathbf{I}_n, \end{aligned} \right. \end{aligned} \quad (12)$$

where $\overline{\mathbf{W}} := \nu\mathbf{W}$ and $\nu \in \mathbb{R}_{>0}$ is a normalizing term of $\mathbf{M}$. Using $\lambda \in \mathbb{R}_{>0}$, the magnitude of the contraction metric $\mathbf{M}$ can be adjusted.

As well known, the computational complexity of solving the above LMI optimization problem is $\mathcal{O}(n^6)$ [...], which

can be a critical issue for high-dimensional systems. In this regard, the conventional method is not suitable for real-time implementation.

Moreover, since CV-STEM penalizes the magnitude of the control input indirectly by adjusting $\mathbf{M}$, the input saturation constraint is not explicitly considered in the optimization problem. Thus, Theorem 1 cannot be directly applied to the system (5) with input saturation, since $\phi(\mathbf{u})$ is not considered as bounded disturbances.

In conclusion, though CV-STEM provides a systematic framework for contraction metric calculation in LMI framework, there still remain several issues to be resolved for practical real-time implementation.

Remark 2. There exist a severe problem formulation issue in the conventional LMI framework. In the formulation, $\frac{d}{dt}\bar{\mathbf{W}} = \frac{d}{dt}(\nu\mathbf{W})$ is considered as same as $\nu\frac{d}{dt}(\mathbf{W})$, which is not mathematically valid since ν is also a decision variable. Thus, the conventional LMI framework's solutions are typically decides $\bar{\mathbf{W}}$ and χ to $\mathbf{I}_n$ and 1, respectively, and modifies ν only.

5. MAIN RESULTS

Since the contraction metric $\mathbf{M}$ is only applied to the trajectory error vector $\mathbf{e} := \mathbf{x} - \mathbf{x}_d$ in the control law (9), we can project the contraction constraints onto the trajectory error vector space. By pre- and post-multiplying $\mathbf{e}^\top$ and $\mathbf{e}$ to the contraction constraint (11), respectively, we have:

$$\mathbf{y}^\top \left(-\mathbf{B}\mathbf{R}^{-1}\mathbf{B}^\top \right) \mathbf{y} + \mathbf{e}^\top \left(\frac{1}{T_s} + 2\alpha + 2\mathbf{A}^\top \right) \mathbf{y} - \frac{\mathbf{e}^\top \mathbf{M}_{\text{pre}} \mathbf{e}}{T_s} \leq 0, \quad (13)$$

where $\mathbf{y} := \mathbf{M}\mathbf{e}$ denotes a projected contraction metric on error vector and a new optimization variable. Note that $\frac{d}{dt}\mathbf{M} \approx \frac{1}{T_s}(\mathbf{M} - \mathbf{M}_{\text{pre}})$ where T_s denotes the sampling time and $\mathbf{M}_{\text{pre}}$ denotes the contraction metric obtained at the previous time step, for discrete-time implementation.

The convex input saturation function $\text{sat}(\cdot)$ preserves the convexity on $\mathbf{y}$, since $\mathbf{u}$ is linear with respect to $\mathbf{y}$. Thus, the input saturation constraint can be expressed as a convex constraint function of $\mathbf{y}$ as follows:

$$c(\mathbf{y}) = \text{sat}(\mathbf{u}) \leq 0. \quad (14)$$

According to Theorem 1, the steady-state error bound is proportional to the condition number of the contraction metric $\mathbf{M}$. We can indirectly minimize the condition number by minimizing the angle between $\mathbf{y}$ and $\mathbf{e}$, since the optimal condition number of $\mathbf{M}$ is achieved when $\mathbf{y}$ is parallel to $\mathbf{e}$, resulting $\mathbf{M} = \gamma\mathbf{I}_n$ for some $\gamma \in \mathbb{R}_{>0}$.

However, there is infinite number of solutions $\mathbf{y}$ which is parallel to $\mathbf{e}$, i.e., since the condition number is scale-invariant. To this end, energy-based constraint is introduced to obtain a unique solution, maintaining sufficient control authority. Specifically, the following energy-based constraint is considered:

$$(\mathbf{e}^\top \mathbf{y} - \mu \mathbf{e}^\top \mathbf{e})^2 \leq (\epsilon + s) \|\mathbf{e}\|^2, \quad (15)$$

where $\mu \in \mathbb{R}_{>0}$ denotes the desired energy level, and $\epsilon \in \mathbb{R}_{>0}$ and $s \in \mathbb{R}_{\geq 0}$ denote allowable small energy variation and a slack variable to enhance the robust feasibility of the optimization problem.

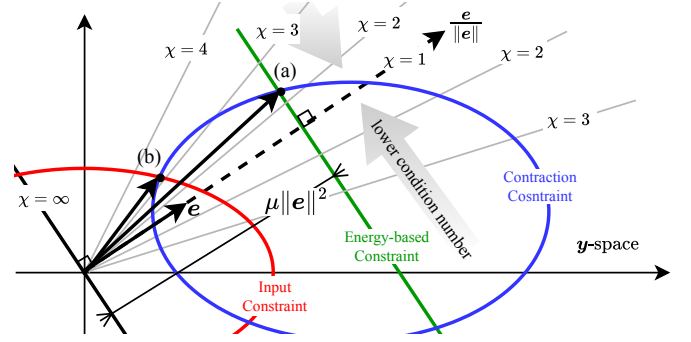


Fig. 1. Illustration of the optimization problem in two-dimensional space.

Thus, the following vector-valued convex optimization problem can be formulated to obtain the projected contraction metric $\mathbf{y}$:

$$\min_{\mathbf{y}, s} \|\mathbf{y}\| \|\mathbf{e}\| - \mathbf{y}^\top \mathbf{e} + \lambda s \quad \text{s.t. (13), (14), (15),} \quad (16)$$

After the optimization, the contraction metric $\mathbf{M}$ can be retrieved using Broyden-Fletcher-Goldfarb-Shanno (BFGS) method as follows:

$$\mathbf{M} = \mathbf{M}_{\text{pre}} - \frac{\mathbf{M}_{\text{pre}} \mathbf{e} \mathbf{e}^\top \mathbf{M}_{\text{pre}}}{\mathbf{e}^\top \mathbf{M}_{\text{pre}} \mathbf{e}} + \frac{\mathbf{y} \mathbf{y}^\top}{\mathbf{e}^\top \mathbf{y}}, \quad (17)$$

which satisfies $\mathbf{M}\mathbf{e} = \mathbf{y}$.

For better intuition, the optimization problem (16) is illustrated in two-dimensional space in Fig. 1. The contraction constraint (13) and the input saturation constraint (14) restrict the feasible space of $\mathbf{y}$ in the outside of blue ellipse and the inside of red ellipse, respectively. The green line represents the energy-based constraint (15) assuming that the allowable energy variation ϵ is approximately zero.

Neglecting the input saturation constraint, the optimal solution is obtained at the point (a) accordingly the objective function. Considering the input saturation constraint, the optimal solution is obtained at the point (b) by increasing the slack variable s . This also ensures that there exists a unique optimal solution in the optimization problem (16).

6. NUMERICAL VALIDATION

6.1 Numerical Validation Set Up

We employed the Lorenz system [MS: REF] to validate the proposed method which is represented as follows:

$$\underbrace{\frac{d}{dt} \begin{pmatrix} x \\ y \\ z \end{pmatrix}}_{=: \mathbf{x}} = \underbrace{\begin{pmatrix} \sigma(y-x) \\ x(\rho-z)-y \\ xy-\beta z \end{pmatrix}}_{=: \mathbf{f}(\mathbf{x})} + \underbrace{\begin{pmatrix} u_x \\ u_y \\ u_z \end{pmatrix}}_{=: \mathbf{u}} + \underbrace{\begin{pmatrix} d_x \\ d_y \\ d_z \end{pmatrix}}_{=: \mathbf{d}}, \quad (18)$$

where $\sigma = 10$, $\rho = 28$, and $\beta = \frac{8}{3}$ are system parameters. The external disturbances $\mathbf{d}$ were modeled as follows:

$$\mathbf{d}(t) := 1000(H(t-0.1) - H(t-0.11)), \quad (19)$$

where $H(\cdot)$ denotes the Heaviside step function.

For the desired trajectory, we simulated the desired system (6) with the following desired feedback-linearization control input:

$$\mathbf{u}_d := -\mathbf{f}_d - 2(\mathbf{x}_d - \mathbf{r}), \quad (20)$$

Table 1. Controllers for comparative study.

Description	
(C ₁) [—]	Proposed controller
(C ₂) [—]	CV-STEM (Tsukamoto and Chung (2021))
(C ₃) [—]	Desired controller, i.e., $\mathbf{u} = \mathbf{u}_d$

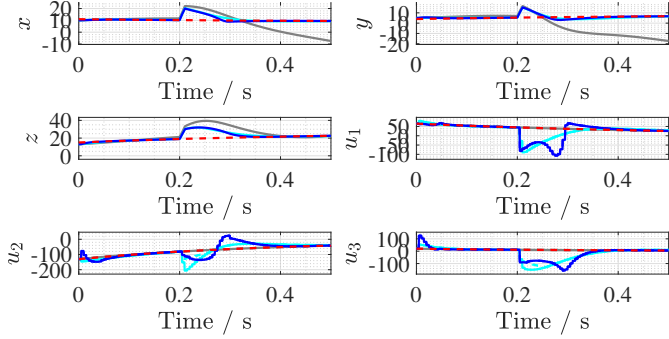


Fig. 2. Numerical validation results of (C₁) [—], (C₂) [—], (C₃) [—], and desired trajectory x_d [—].

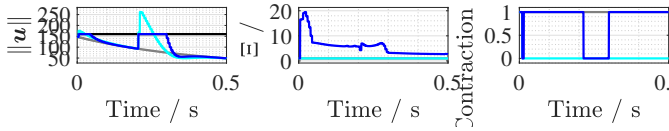


Fig. 3. Optimization results of (C₁) [—] and (C₂) [—].

where $\mathbf{r} := (-8.485, 8.485, 27)^\top$ denotes the reference point.

The proposed method (C₁) was compared with CV-STEM (C₂) Tsukamoto and Chung (2021) and the desired controller $\mathbf{u} = \mathbf{u}_d$ (C₃) as summarized in Table 1. The parameters shared by (C₁) and (C₂) were $\mathbf{R} = \mathbf{I}_3$ and $\alpha = 10$. In addition, parameters were selected as $\mu = 30$ and $\lambda = 100$ for (C₁) and $\lambda = 0.01$ for (C₂). To solve the optimization problems, `fmincon` function in MATLAB R2024a and `SeMeDi` were used for (C₁) and (C₂), respectively.

6.2 Numerical Validation Results

7. CONCLUSION

In this study, ...

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Place acknowledgments here.

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